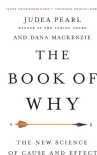
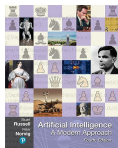


Lesson 08.3: Do Calculus

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References:

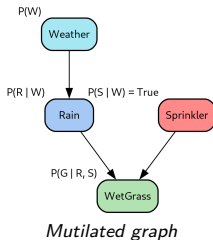
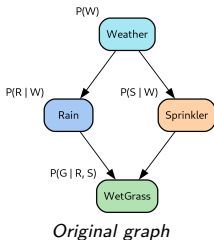
- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017



- *Intervention and Counterfactuals*
- Randomized Controlled Trial
- Back-door Adjustment
- Front-door Adjustment
- Do-Calculus

Interventions in Causal Networks

- **Causal Bayesian Networks** represent cause–effect relations between variables
 - E.g., $Rain \rightarrow WetGrass$
- **Interventions**
 - *Intervention* means setting a variable to a fixed value, overriding its causal mechanism
- E.g., “Turning the sprinkler on manually” regardless of cloudiness
 - Replace equation $S = f_S(C, U_S)$ with $S = true$
 - Causal link from *Cloudy* to *Sprinkler* is *cut*, forming a new “mutilated” model



Interventions in Causal Networks

- **The do-operator**

- Denoted as $\text{do}(X = x)$
- Represents performing an action that *sets* X to x , not *observing* $X = x$
- $\text{do}(X_j = x_j^k)$ removes $\Pr(x_j | \text{parents}(X_j))$ from the product and gives a new joint distribution:

$$P_{X_j=x_j^k}(x_1, \dots, x_n) = \begin{cases} \prod_{i \neq j} \Pr(x_i | \text{parents}(X_i)) & \text{if } X_j = x_j^k \\ 0 & \text{otherwise} \end{cases}$$

- **Difference between observation and intervention**

$$\Pr(Y | \text{do}(X = x)) \neq \Pr(Y | X)$$

- Observing $S = \text{true}$ *provides information* about its causes
 - * E.g., information about weather (and the Markov blanket)
- Intervening with $\text{do}(S = \text{true})$ *breaks* those causal dependencies
 - * E.g., it doesn't inform about anything

Intervention

- **Estimate causal effect** of X_j on X_i with adjustment formula:

$$\Pr(X_i = x_i | \text{do}(X_j = x_j^k)) = \sum_{\text{parents}(X_j)} \Pr(x_i | x_j^k, \text{parents}(X_j)) \Pr(\text{parents}(X_j))$$

- **Example**

- In the Sprinkler model $\text{do}(S = \text{true})$, gives the new distribution:

$$\Pr(c, r, w, g | \text{do}(S = \text{true})) = \Pr(c) \Pr(r|c) \Pr(w|r, S = \text{true}) \Pr(g|w)$$

- * Only descendants of *Sprinkler* (i.e., *WetGrass*) change
- * *Weather* and *Rain* remain unaffected

- **Intuition**

- Do-operator isolates *causal effects* by simulating external manipulation
- Essential for answering “what if” questions: *What happens if you intervene and change X?*

Counterfactuals

- A **counterfactual** describes what would have happened in the past under a different scenario
 - *“What would the outcome have been, if X had been different?”*
- **Business examples**
 - *“What if we had two suppliers instead of one? Would we have fewer delays?”*
 - *“Would customers be more satisfied if we shipped products in one week instead of three?”*
 - That’s what businesses want, but they can’t get it from correlation-based models!
- **Causal reasoning**
 - Goes beyond correlation and association
 - Requires a causal model to simulate alternate realities
 - E.g.,
 - * Actual: *“A student received tutoring and scored 85%”*
 - * Counterfactual: *“What if the student didn’t receive tutoring?”*
 - * Causal model estimates the alternative outcome (e.g., 70%)
- **Challenges**
 - Requires strong assumptions and accurate models
 - Difficult to validate directly since counterfactuals are unobservable

Causal Discovery

- **Definition**

- Causal discovery learns causal network structure from data
- Identify which variables directly cause others (learn causal directions, not just correlations)

- **Approaches to causal discovery**

- **Search-based methods**

- * Start with an empty or initial model and iteratively modify it (add, reverse, or delete links)
- * Evaluate each candidate network based on fit to data (e.g., likelihood)
- * Use search strategies while ensuring the network remains acyclic

- **Constraint-based methods**

- * Infer causal directions from conditional independence tests among variables
- * If X and Y are independent given Z , this constrains possible arrows

- **Dealing with complexity**

- Possible network structures grow super-exponentially with the number of variables
- Complexity penalties to avoid overfitting

- Intervention and Counterfactuals
- ***Randomized Controlled Trial***
- Back-door Adjustment
- Front-door Adjustment
- Do-Calculus

What is a Randomized Controlled Trial?

- **RCTs estimate causal effects** by comparing treatment and control groups
 - Randomly assign treatment, not chosen by subjects
 - * E.g., assign new drug vs placebo by flipping a coin
 - Ensure groups are statistically equivalent except for treatment
 - Isolate treatment effect
- **Goal:** estimate $\Pr(Y|do(X))$
 - Randomization simulates do-operator by removing incoming arrows to X
 - Eliminate (known and unknown) confounding paths from background variables
 - Turn observational data into experimental data
 - * Use $\Pr(Y|X)$ to measure $\Pr(Y|do(X))$
 - Allow causal inference without knowing the causal graph
- **Pros**
 - Implement intervention in a principled, unbiased way
 - Foundation for scientific experimentation and evidence-based policy
 - Gold standard of causal inference (when feasible)

Randomized Controlled Trial: Example

- **Research question:** *“Does offering an after-school tutoring program increase the probability that a student passes the end-of-term exam?”*
- **Population:** eligible students in a district with proper sample size n
- **Treatment and control**
 - $X = 1$ (treatment): student is offered/assigned to tutoring
 - $X = 0$ (control): student is not offered tutoring
- **Assignment mechanism (RCT)**
 - Students are randomly assigned to $X \in \{0, 1\}$
 - Randomization ensures, in expectation, balance on prior GPA, motivation, parental income, etc
- **Outcome**
 - $\Pr(Y \mid do(X)) = \Pr(Y \mid X)$
 - $Y_{X=x} = I\{\text{pass exam}\}$ measured at term's end for treatment vs control
 - Measure and report $Y_{X=1} - Y_{X=0}$

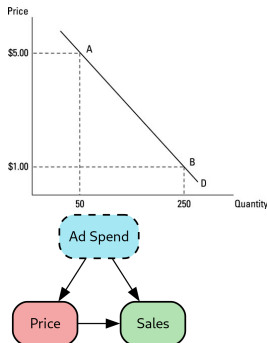
Randomized Controlled Trial: Limits

- **May be unethical**
 - E.g., assigning harmful treatment
 - E.g., you want to verify if asbestos causes cancer
- **Can be expensive or impractical**
- **It doesn't always work**
 - Non-compliance: some participants may not follow assigned treatment
 - Attrition: dropout rates may differ between groups
 - May not generalize to broader populations
 - Requires careful implementation and monitoring
- **Blind RCT**: participants don't know which group they're in (e.g., placebo)
- **Double-blind RCTs**: participants and investigators/clinicians don't know assignments

- Intervention and Counterfactuals
- Randomized Controlled Trial
- ***Back-door Adjustment***
- Front-door Adjustment
- Do-Calculus

Back-Door Paths: Example

- The company **needs to control** for *AdSpend* to estimate the causal effect of *Price* on *Sales* by:
 - Using *AdSpend* as covariate in the regression
 - Designing experiment holding *AdSpend* constant or randomized
 - Using back-door criterion
- **Example**
 - A company wants to understand the causal effect of *price* on *sales*
 - Advertising spend *AdSpend* is a **confounder** since it can affect both:
 - * The *price* the company can set
 - * E.g., the cost increases to cover advertisement costs and the product is perceived as more valuable
 - * The *sales* (directly)
- The **back-door path** is $Price \rightarrow AdSpend \rightarrow Sales$



The Back-Door Adjustment

- **Hypotheses**

- You have a (correct) causal graph
- You block the back-door paths (needs definition!) that satisfy the back-door criterion (needs definitions!)

- **Thesis**

- The *adjustment formula* holds

$$\Pr(Y \mid do(X)) = \sum_z \Pr(Y \mid X, Z = z) \Pr(Z = z)$$

- **Consequences**

- It allows you an intervention (level 2 of the causality ladder) only using observational data (level 1 of the causality ladder)
 - * Correlation implies causation
- This is an alternative to randomized controlled experiments
- **Mind blown!**

Back-Door Criterion: Overview

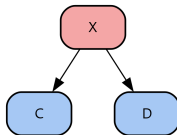
- A **back-door path** is any path from X to Y starting with an arrow into X and ending pointing into Y
 - E.g., $X \leftrightarrow \dots \leftrightarrow \dots \rightarrow Y$
 - Arrow direction doesn't matter
 - Unblocked paths create spurious associations
 - *Intuition*: Paths make X and Y look related even if changing X doesn't change Y
- **Back-door criterion**: A set of variables Z satisfies the criterion relative to $X \leftrightarrow \dots Z \dots \rightarrow Y$ if:
 - No variable in Z is a descendant of X
 - Z blocks every path from X to Y starting with an arrow into X
 - *Intuition*: Block variables that interfere with the effect you're estimating and remove paths where X and Y connect through common causes
- Condition on variables satisfying the back-door criterion to estimate a causal effect without intervention
 - This guy won the Nobel prize for CS for this! **Mind blown!**

Chains, Forks, and Colliders

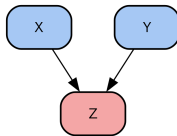
- In a **chain** $X \rightarrow M \rightarrow Y$
 - Conditioning on M blocks causal effect
 - Mediators must remain unconditioned
 - 🦴 *Do not do it!*



- In a **fork** $X \leftarrow Z \rightarrow Y$
 - Conditioning on Z removes confounding
 - 😊 *Need to do it!*



- In a **collider** $X \rightarrow M \leftarrow Y$
 - Conditioning on M introduces bias
 - Colliders must remain unconditioned
 - 🦴 *do not do it!*



Common Mistakes

- The back-door criterion tells you all and only what you need to condition on (i.e., block) to:
 - Transform observation in intervention
 - Say that “correlation IS causation”
- Before this, the solution that researchers used was to “condition on everything”
 - **This is incorrect!**
- **Common mistakes**
 - Conditioning on a descendant of X can bias the estimate
 - Controlling for too many variables can open colliders and introduce bias
 - Forgetting to block all back-door paths
 - Using variables that lie on the causal path (blocks the mediator of the effect)
 - Ignoring unobserved confounders: can make causal effect unidentifiable
- **1000s of papers and their conclusions are wrong!**
 - In medicine, economics, social science people have used observational study incorrectly

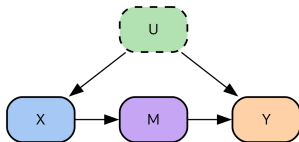
When Back-Door Adjustment Fails

- **Back-door is simple but not universally applicable**
 - No set of observable variables satisfies the back-door criterion
 - * In order to condition on a variable it needs to be observable
 - * E.g., an unobserved or unknown confounders
 - You need to know a good approximation of the true causal graph
 - * Very easy to omit variables
- **Alternatives:**
 - *Front-door criterion*: uses mediators
 - *Instrumental variables*: uses external variation
 - *Do-calculus*: symbolic transformations to eliminate $do()$

- Intervention and Counterfactuals
- Randomized Controlled Trial
- Back-door Adjustment
- *Front-door Adjustment*
- Do-Calculus

Front-Door Adjustment in Causal Inference

- **Front-door criterion** identifies causal effects with unobserved confounders
 - Applies when a **mediator variable** transmits all causal influence from treatment to outcome
- Assume the causal graph looks like:
 - X : treatment or cause
 - M : mediator
 - Y : outcome
 - U : unobserved confounder
- **Hypotheses:**
 - All directed paths from X to Y go through M
 - No unobserved confounder affects X and M
 - All backdoor paths from M to Y are blocked by X
- **Thesis:** estimate the causal effect $P(Y|do(X))$ despite unobserved U

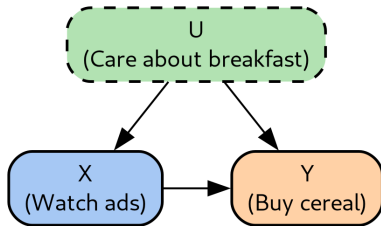


$$P(Y | do(X)) = \sum_m P(M | X) \sum_{x'} P(Y | M, X') P(X')$$

- *Intuition:* estimate observed link $X \rightarrow M$ and $M \rightarrow Y$

Cereal and Ads: Example

- **Research question:** “Does watching ads (X) make people buy more cereal (Y)?”
- You might think **yes, of course!**, I say **no, not so fast!**
- **Hidden factor:** “Parents who care about breakfast (U)” might:
 - Let kids watch more TV to have them eat breakfast and see ads
 - Buy more cereal anyway
- Hidden factor U confounds “Watch ads” and “Buy cereals”
 - Correlation exists even if ads don’t cause it
 - Observing X and Y without controlling for U leads to spurious association
 - Same of “ice cream” and “drowning”
- A spurious relationship is **terrible for the business!**
 - It means you spend money on ads and that doesn’t matter
 - Google and Facebook are worth \$3T and it’s all predicated on “Watch ads” and “Buy cereal”



Cereal and Ads: Solutions

- **Strategy 1: Back-door adjustment**

- If you *know* and can *measure* U “how much parents care about breakfast”, include U as a control variable in analysis
- *Intuition*:
 - * Compare families with *the same* breakfast attitudes (U fixed)
 - * See if ads (X) still change cereal buying (Y)

- **Strategy 2: Use randomization**

- Randomized experiments break link between X and U
 - * Randomly show ads to some families and not others
 - * Randomization ignores parental breakfast attitude; differences in buying come from ads
- This is why controlled experiments are gold standard for causal inference

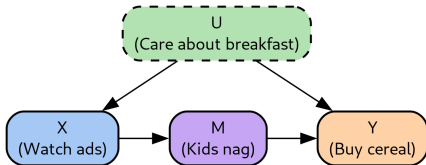
- **Strategy 3: Front-door Adjustment**

Cereal and Ads: Finding a Mediator

- Imagine ads work by “making kids ask for cereal” (aka “nagging”) M
 - This is a true advertisement strategy!
 - At the convenience store the candies are at the bottom of the desk
- There is a **mediator** variable



- So the **causal chain** is:
 - Ads (X) $\rightarrow$...
 - Kids Nagging (M) $\rightarrow$...
 - Parents Buy Cereal (Y)
- The hidden factor “parents that care about breakfast” U :
 - Affects how much cereal gets bought
 - Doesn’t affect how much kids nag (only ads do that)

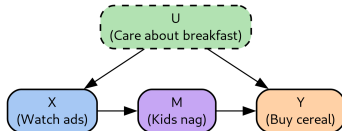


When Front-Door Works

- **Is the front-door criterion verified**

for “kids’ nagging” M ?

- Influence of ads on buying goes through nagging ($X \rightarrow M \rightarrow Y$)
- No hidden confounders affect both ads and nagging (TV schedule is random, not linked to parents’ breakfast attitudes)
- All confounding between nagging and buying is blocked by controlling for ads



- **Yes! The front-door criterion is verified**

- Instead of doing an intervention $do(X)$, just observe!

- Observe how often ads make kids nag, $\Pr(M|X)$
- Observe nagging changes buying, $\Pr(Y|M, X')$
- Combine both to estimate what happens if you *force* more ads

$$\Pr(Y|do(X)) = \sum_m \Pr(M|X) \sum_{x'} \Pr(Y|M, X') \Pr(X')$$

- Intervention and Counterfactuals
- Randomized Controlled Trial
- Back-door Adjustment
- Front-door Adjustment
- ***Do-Calculus***

Do-Calculus

- **Do-calculus** is a formal system for reasoning about causal effects in graphical models (Judea Pearl, 2000)

- **Problem**

- You care about **causal effects** like:

$$\Pr(Y|do(X = x))$$

- * I.e., distribution of Y if you intervene and set X to x , breaking causal links into X

- You have **observational data** like:

$$\Pr(Y|X = x)$$

- In general $\Pr(Y|X) \neq \Pr(Y|do(X = x))$ due to “correlation is not causation” (confounding, ...)

- **Solution**: do-calculus provides algebraic rules to transform intervention expressions (do-operator, i.e., $do(X = x)$) into expressions computable from observational data, given certain conditions

The Rules of Do-Calculus

- Do-calculus provides **three transformation rules** for manipulating expressions involving $do()$:

- Insertion/Deletion of Observations:** If $Y \perp Z \mid X, W$ in $G_{\overline{X}}$ (where incoming edges to X are removed), then:

$$\Pr(Y \mid do(X), Z, W) = \Pr(Y \mid do(X), W)$$

- Action/Observation Exchange:** If $Y \perp Z \mid X, W$ in $G_{\overline{X}, \underline{Z}}$ (incoming edges to X removed, outgoing from Z removed), then:

$$\Pr(Y \mid do(X), do(Z), W) = \Pr(Y \mid do(X), Z, W)$$

- Insertion/Deletion of Actions:** If $Y \perp Z \mid X, W$ in $G_{\overline{X}, \overline{Z(W)}}$ (incoming edges to X and to Z excluding those from W removed), then:

$$\Pr(Y \mid do(X), do(Z), W) = \Pr(Y \mid do(X), W)$$

- These rules allow the systematic reduction of expressions involving $do()$ into observational terms, if the causal graph permits

Back/Front-door Adjustments and Do-calculus

- The **back-door** and **front-door** criteria are **specific applications** of do-calculus
 - They are simpler, graphical conditions that allow $P(Y \mid do(X))$ to be expressed using observational probabilities
- **Back-door adjustment**: If a set of variables Z blocks all back-door paths from X to Y (paths that go into X), then:

$$\Pr(Y \mid do(X)) = \sum_z \Pr(Y \mid X, Z) \Pr(Z)$$

- **Front-door adjustment**: If there exists a variable Z such that:
 - Z is affected by X
 - Z affects Y
 - All back-door paths from X to Z are blocked
 - All back-door paths from Z to Y are blocked by X

then:

$$\Pr(Y \mid do(X)) = \sum_z \Pr(Z \mid X) \sum_{x'} \Pr(Y \mid Z, X') \Pr(X')$$